

Class-II translation identity and translated-current model reduction

TECT verification-first repository

claim A13-CLASSII-RELATIVE-PHASE-SOURCE-BUDGET-OBSTRUCTION

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Purpose and scope

A genuine reduction is closed, but the Nelson theorem is not. For the hash-pinned A1 production coefficients and every finite common-even spectral cutoff, this note proves the exact Class-II Cameron–Martin translation identity, its Pauli-current frame form, deterministic-shift expectation positivity, and a Cartan identity which preserves the antisymmetric curvature of the non-exact K current. It also removes the unnecessarily restrictive half-sextic budget: the sufficient one-use range is $\epsilon_6 < \gamma/6$, not $\epsilon_6 < \gamma/12$.

These results do not prove the cutoff-uniform Nelson moment. The translated random-coefficient terms require a larger enhanced model than A7 states, and an L^3 model-norm moment appears already at the endpoint of the simplest Sobolev estimate. A homogeneous fixture refutes only a cutoff-independent deterministic constant remainder; it does not refute an integrable random remainder. A13 therefore remains T4 and the umbrella gate A13-CLASSII-CONTROLLED-SHELL-ENERGY-ONE-USE remains open.

The scope is the fixed $L = 16$ torus, three complex components in the six-real convention, the A1 production covariance and coefficients, fixed $\rho_{\text{floor}} = 10^{-12}$, common real-even sharp spectral cutoffs, and exact covariance-normal subtraction from A7. All finite-cutoff identities hold for arbitrary fields in the cutoff space. The executable manufactured fields are algebraic regression tests, not continuum convergence evidence.

1. Production definitions

Write the complex field as a six-real vector $X \in \mathbb{R}^6$. For the three embedded Pauli generators S_A , put

$$\rho = X^T X, \quad m_A = X^T S_A X, \quad q_A = \frac{m_A}{\rho + \epsilon_\rho}, \quad (1.1)$$

$$p_A = 2S_A X, \quad v_A = 2(S_A - q_A I)X. \quad (1.2)$$

The production coefficient matrix is

$$B(X) = \sum_A M_A(X) Q_{\text{II}} M_A(X)^T, \quad M_A = [p_A, v_A], \quad Q_{\text{II}} = \begin{pmatrix} a & b \\ b & c \end{pmatrix}. \quad (1.3)$$

Here

$$a = 0.004499999999998875, \quad b = 0.0018749999999995311, \quad c = 0.002343749999999414, \quad (1.4)$$

and the eigenvalues of Q_{II} are

$$0.00125901150092606, \quad 0.00558473849907223. \quad (1.5)$$

Consequently $B(X) \geq 0$ for every X , including the floor region.

For $Y_i = \partial_i X$ and the exact six-real derivative covariance $\Gamma_{J,i}$, A7 defines

$$V_J^{\text{ren}}(X) = \frac{1}{2} \sum_i \int_{\mathbb{T}_L^3} \left\{ Y_i^T B(X) Y_i - \text{Tr}[B(X) \Gamma_{J,i}] \right\} dx. \quad (1.6)$$

The subtraction in (1.6) is regulator dependent through the exact $\Gamma_{J,i}$; a leading asymptotic scalar is not substituted for it.

2. Flexible production-potential coercivity

The local scalar potential is

$$U(r) = \frac{\lambda}{4} r^2 + \frac{\gamma}{6} r^3, \quad r = |z|^2, \quad \lambda = -0.43, \quad \gamma = 1.62. \quad (2.1)$$

Lemma 2.1. For every $\delta > 0$ and every $r \geq 0$,

$$U(r) \geq \left(\frac{\gamma}{6} - \delta \right) r^3 - \frac{|\lambda|^3}{432\delta^2}. \quad (2.2)$$

Proof. Put $A = |\lambda|/4$. The minimum of $\delta r^3 - A r^2$ occurs at

$$r_* = \frac{2A}{3\delta} = \frac{|\lambda|}{6\delta}, \quad (2.3)$$

and its value is $-|\lambda|^3/(432\delta^2)$. Adding $(\gamma/6 - \delta)r^3$ proves (2.2). Equality holds in the auxiliary inequality at (2.3). $\square$

The production stress choice used in this package is

$$\epsilon_6 = 0.15, \quad \delta = 0.06, \quad \frac{\gamma}{6} - \delta - \epsilon_6 = 0.06. \quad (2.4)$$

The constant density and the $L = 16$ finite-volume constant are

$$\frac{|\lambda|^3}{432\delta^2} = 0.0511233281893004\dots, \quad L^3 \frac{|\lambda|^3}{432\delta^2} = 209.401152263374\dots \quad (2.5)$$

Thus any one-use estimate with field charge $\epsilon_6 < \gamma/6 = 0.27$ leaves a coercive sextic after choosing a sufficiently small positive $\delta < \gamma/6 - \epsilon_6$. The previous $\gamma/12 = 0.135$ requirement was a convenient equal split, not the sharp sufficient range. This correction is registered as **AUDIT-2026-07-22-A13-HALF-SEXTIC-OVERRESTRICTION**.

At $p = 1.1$, the retained control candidate is

$$\epsilon_v = 0.45 < \frac{1}{2p} = 0.454545\dots \quad (2.6)$$

The separated direct Ramer-square carrier charge 0.14136616176932618 now fits below 0.15. This does not revive the one-shot Ramer map: the v1.1 production determinant crosses zero at the correct displacement coefficient 5/9. The surviving obstruction is the fold, not the old half-sextic comparison.

3. Exact finite-cutoff translation identity

Let h belong to the same finite cutoff space and set

$$H_i = \partial_i h, \quad B_+ = B(X + h), \quad \Delta B = B_+ - B(X). \quad (3.1)$$

Theorem 3.1. At every finite cutoff,

$$\begin{aligned} V_J^{\text{ren}}(X + h) - V_J^{\text{ren}}(X) &= \frac{1}{2} \sum_i \int \Delta B : (Y_i \otimes Y_i - \Gamma_{J,i}) \, dx \\ &\quad + \sum_i \int H_i^T B_+ Y_i \, dx + \frac{1}{2} \sum_i \int H_i^T B_+ H_i \, dx. \end{aligned} \quad (3.2)$$

Proof. Insert $Y_i + H_i$ and B_+ into (1.6), subtract the expression at X , and expand the quadratic form. Symmetry of B_+ combines the two mixed terms. The counterterm changes only by $\text{Tr}(\Delta B \Gamma_{J,i})$, because a deterministic translation does not change the Gaussian covariance. $\square$

The final line of (3.2) is nonnegative. The first two terms are signed and must remain coupled. Bounding them separately discards precisely the centered covariance cancellation fixed in A7 and A9.

There is a second exact form. Put

$$w_{A,i} = M_A(X)^T Y_i, \quad \Delta w_{A,i} = M_A(X + h)^T (Y_i + H_i) - w_{A,i}, \quad \Delta M_A = M_A(X + h) - M_A(X). \quad (3.3)$$

Then

$$\begin{aligned} \Delta V_J^{\text{ren}} &= \sum_{A,i} \int \{ w_{A,i}^T Q_{\Pi} \Delta w_{A,i} + \frac{1}{2} \Delta w_{A,i}^T Q_{\Pi} \Delta w_{A,i} \\ &\quad - \text{Tr}(Q_{\Pi} M_A^T \Gamma_{J,i} \Delta M_A) - \frac{1}{2} \text{Tr}(Q_{\Pi} \Delta M_A^T \Gamma_{J,i} \Delta M_A) \} dx. \end{aligned} \quad (3.4)$$

The Δw square is positive, while the last counterterm square is negative. Formulae (3.2) and (3.4) are alternative decompositions of the same quantity; their positive squares must not be counted twice.

4. Deterministic translations are positive in expectation

For a common real-even regulator,

$$\mathbb{E}[X(x) \otimes \partial_i X(x)] = 0. \quad (4.1)$$

Joint Gaussianity therefore makes $X(x)$ and $Y_i(x)$ independent. If h is deterministic, $B(X + h)$ and ΔB depend only on X . Conditioning on X in (3.2) gives

$$\mathbb{E}[\Delta B : (Y_i \otimes Y_i - \Gamma_{J,i}) \mid X] = 0, \quad \mathbb{E}[H_i^T B_+ Y_i \mid X] = 0. \quad (4.2)$$

Hence

$$\mathbb{E}\{V_J^{\text{ren}}(X + h) - V_J^{\text{ren}}(X)\} = \frac{1}{2} \sum_i \int H_i^T \mathbb{E}[B(X + h)] H_i \, dx \geq 0. \quad (4.3)$$

The executable checks (4.3) by exact covariance-reproducing sigma points and an independent Gauss–Hermite rule. Equation (4.3) does not apply to an adapted random $h(X)$; that dependence is the remaining problem.

5. Cartan identity for the non-exact current

For each Pauli generator define the target-space one-form

$$\omega_A = dm_A - q_A d\rho, \quad K_A(X) = X^* \omega_A. \quad (5.1)$$

It is not exact. Direct differentiation gives

$$d\omega_A = \frac{1}{\rho + \epsilon_\rho} d\rho \wedge dm_A, \quad \|d\omega_A\|_{\text{op}} \leq \frac{4\rho}{\rho + \epsilon_\rho} \leq 4. \quad (5.2)$$

Let $X_t = X + th$. Cartan's homotopy formula along this straight target-space path gives the exact spatial-current translation

$$\begin{aligned} K_A(X + h) - K_A(X) &= d_x \int_0^1 \omega_A(X_t)[h] dt \\ &\quad + \int_0^1 d\omega_A(X_t)[h, d_x X_t] dt. \end{aligned} \quad (5.3)$$

Proof. Differentiate $X_t^* \omega_A$ in t . In coordinates the result is $D\omega_A(X_t)[h]d_x X_t + \omega_A(X_t)d_x h$. Add and subtract $D\omega_A(X_t)[d_x X_t]h$; the exact derivative becomes $d_x \{\omega_A(X_t)[h]\}$ and the remainder is $d\omega_A(X_t)[h, d_x X_t]$. Integrate from zero to one. $\square$

Formula (5.3) is stronger than a direct termwise Taylor estimate of B : the rough dependence remains linear in $h dX$, and the nonzero Class-II curl from v1.1 is retained with its signed antisymmetry.

6. Why a deterministic constant remainder is impossible

Consider the stronger, pathwise proposal with a cutoff-independent constant C ,

$$\Delta V_J^{\text{ren}}(X; h) \geq -\eta \|h\|_{\mathcal{H}_C}^2 - \epsilon_6 \|X + h\|_6^6 - C. \quad (6.1)$$

At the Gaussian-null point $X = 0$, choose a constant active-doublet translation $h = te_1$. Its currents vanish, so

$$V_J^{\text{ren}}(te_1) = -C_J(te_1). \quad (6.2)$$

The A7 sharp-cube asymptotic is

$$\frac{C_J(te_1)}{Jt^2 L^3} \longrightarrow \delta_{\text{cube}}(9a + 12b + 6c) = 0.00124833439336115\dots \quad (6.3)$$

The Cameron–Martin cost is only $O(t^2)$ with a J -independent coefficient. After that coefficient changes sign, minimising $-A_J t^2 + \epsilon_6 t^6$ yields a deficit proportional to $A_J^{3/2} = \Theta(J^{3/2})$. Thus (6.1) is false.

This is a deliberately narrow no-go. The point $X = 0$ has Gaussian measure zero, and a random remainder $R_J(X)$ can be large there while satisfying $\sup_J \mathbb{E} R_J(X) < \infty$. The calculation does not refute the canonical expected adapted-control inequality.

7. The exact translated-model gap

For the production covariance $C(k) \asymp \langle k \rangle^{-4}$ in three dimensions, take any

$$\frac{1}{3} < \alpha < \frac{1}{2}, \quad \kappa > 0. \quad (7.1)$$

A7 supplies the particular contracted scalar $B(X) : (\partial X \otimes \partial X - \Gamma)$ in $L^2(\Omega; H^{-1-\kappa})$. It does not state the universal tensor or the coefficient jets needed after translation. The smallest non-circular next theorem is

$$\text{A13-CLASSII-CAMERON-MARTIN-TRANSLATED-CURRENT-MODEL-LIFT.} \quad (7.2)$$

It must construct, consistently across cutoffs and with all required finite moments,

model object	target regularity	
X	C^α	
$\mathbb{X}_i^{ab} = X^a \circ \partial_i X^b$	$C^{2\alpha-1}$	
$Q_i^{ab} =: \partial_i X_a \partial_i X_b :$	$H^{-1-\kappa}$	
$X^c \diamond Q_i^{ab}$	$H^{\alpha-1-\kappa}$	
$: X^c X^d : \diamond Q_i^{ab}$	$H^{2\alpha-1-\kappa}$	(7.3)

The derivative-pair covariance $\Gamma_{J,i}$ is the only local subtraction allowed. Common evenness makes the same-point value-derivative contraction zero. Any new divergent interaction counterterm would be a new obstruction, not an admissible silent repair.

The moment requirement is already visible in one deterministic Sobolev calculation. Testing Q against h^2 gives

$$\|h^2\|_{H^{1+\kappa}} \lesssim \|h\|_{H^2}^{1+\kappa/2} \|h\|_6^{1-\kappa/2}. \quad (7.4)$$

Three-factor Young therefore requires the model norm of Q in

$$L^{6/(2-\kappa)}(\Omega), \quad (7.5)$$

which tends to L^3 as $\kappa \downarrow 0$. A7's stated L^2 result does not imply (7.5).

There is also a critical signed-current boundary. Termwise absolute-value estimates produce schematically

$$|\langle \mathbb{J}, h^2 \nabla h \rangle| \lesssim \|\mathbb{J}\|_{H^{-1/2}} \|h\|_{H^2}^{3/2} \|h\|_6^{3/2}. \quad (7.6)$$

Since

$$\frac{3/2}{2} + \frac{3/2}{6} = 1, \quad (7.7)$$

Young's inequality leaves no exponent with which to absorb an unbounded random $\|\mathbb{J}\|$. The positive current square, centered Q , and the Cartan curvature term must therefore be estimated together. Generic termwise paraproduct absorption is not a valid successor architecture.

8. Conditional route to the umbrella one-use theorem

For fixed $\eta, \epsilon_6 > 0$, define the strongest useful stochastic form deficit

$$R_J^*(X) = \sup_h \left[-\Delta V_J^{\text{ren}}(X; h) - \eta \|h\|_{\mathcal{H}_C}^2 - \epsilon_6 \|X + h\|_6^6 \right]_+. \quad (8.1)$$

If the translated model lift and a signed reconstruction theorem prove

$$\sup_J \mathbb{E} R_J^*(X) < \infty, \quad (8.2)$$

then substituting the adapted endpoint shift $h = h(v)$ and using $\|h(v)\|_{\mathcal{H}_C}^2 \leq \mathcal{E}(v)$ gives the required one-use estimate. The identity $\mathbb{E} V_J^{\text{ren}}(X) = 0$ supplies the unshifted term. Combining that estimate with Lemma 2.1 yields the $q = 10/9$ Nelson moment.

Equation (8.2) may be stronger than necessary. If the all- h supremum is too strong, the honest fallback is the strict-past shell-progressive version for triangular adapted shifts. Merely renaming the original Nelson moment as a Bellman value is circular and remains forbidden by v1.1.

9. Executable audit

The primary implementation constructs B from the six-real Pauli frames and checks (3.2), (3.4), and (5.3). Its field matrix contains zero, homogeneous, band-limited random, exact q_0 -shell, Class-II relative-phase, high-frequency, pure-third, and near-floor cases. Exact sharp-cube covariances reproduce the deterministic-remainder no-go.

The independent implementation does not import the primary module. It computes J_A and K_A directly in complex notation, uses an independently implemented Fourier mode-count convolution, checks the endpoint difference by path quadrature, and verifies (4.3) with tensor Gauss–Hermite quadrature. The integrated verifier pins every source, authority, note, and PDF hash and checks the claim, gate, roadmap, result, task, and negative-result records.

The operator command is

```
python codes/foundations/a13_classii_translation_model_reduction_verify.py
```

Expected verdict:

A13-CLASSII-TRANSLATION-MODEL-REDUCTION-INTEGRATED-PASS

Devil’s-advocate

1. **DISMISSED: the old $\gamma/12$ condition is necessary.** Lemma 2.1 keeps the quartic absorption parameter independent of the one-use field charge. For every $\epsilon_6 < \gamma/6$, a positive $\delta < \gamma/6 - \epsilon_6$ leaves a sextic margin and only a finite-volume constant.
2. **VALID WITH MITIGATION: fitting the Ramer square below 0.15 revives the one-shot map.** It does not. The immutable v1.1 finite-mode fixture has a zero and sign change of $\det(I + (5/9)Db_J)$; this package reclassifies the square comparison but preserves the determinant no-go.
3. **DISMISSED: the positive final term in (3.2) permits separate absolute estimates of the first two terms.** The frame identity (3.4) shows a negative translated-counterterm square of the same order. Separating the terms loses the centered Gaussian cancellation and can manufacture a false coercivity statement.
4. **VALID WITH MITIGATION: deterministic expectation positivity proves the adapted case.** It does not. Formula (4.2) uses independence of h from X and Y . The remaining theorem is explicitly restricted to random/adapted translations and requires the new model lift.
5. **DISMISSED: the homogeneous fixture refutes every random remainder.** It refutes only a cutoff-independent deterministic constant. The Gaussian-null evaluation at $X = 0$ supplies no lower bound on $\mathbb{E}R_J(X)$.
6. **VALID WITH MITIGATION: A7’s L^2 composite already contains all tensors in (7.3).** The A7 theorem states one coefficient-contracted scalar. Decontraction and translated coefficient jets are not consequences of that statement. The next claim must construct them and pin the absence of new counterterms.
7. **UPHELD AGAINST TERMWISE METHODS: ordinary paraproduct Young estimates close the last gate.** The endpoint cubic moment in (7.5) and the saturated exponents in (7.7) show that such an argument has no random-norm absorption slot. Only a coupled signed reconstruction remains viable.

Result footer

Claim: A13-CLASSII-RELATIVE-PHASE-SOURCE-BUDGET-OBSTRUCTION

Result: A13-CLASSII-TRANSLATION-MODEL-REDUCTION

Tier: T4 (unchanged)

Closed: flexible potential coercivity; exact finite-cutoff translation and frame identities; deterministic-shift expectation positivity; Cartan current identity; deterministic-constant remainder no-go; exact translated-model and moment boundary.

Open: A13-CLASSII-CONTROLLED-SHELL-ENERGY-ONE-USE.

Next: A13-CLASSII-CAMERON-MARTIN-TRANSLATED-CURRENT-MODEL-LIFT, followed by a coupled centered-Q/current/curvature stochastic form bound.

Honesty: this result does not prove the cutoff-uniform Nelson moment, a random/adapted translation estimate, an interacting measure, floor or regulator removal, infinite volume, phase transition, BCC, T5, T6, or T7.